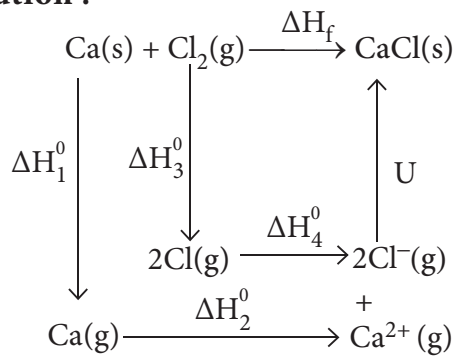
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64.

Solution :



$\Delta H_f = \Delta H_1 + \Delta H_2 + \Delta H_3 + 2\,\Delta H_4 + u$

$-795 = 121 + 2422 + 242.8 + (2 \times -355) + u$

$-795 = 2785.8 - 710 + u$

$-795 = 2075.8 + u$

$u = -795 - 2075.8$

$u = -2870.8 \text{ kJ mol}^{-1}$

65.

Solution :

Given :

$\Delta H_f(Fe_2O_3) = -741 \text{ kJ mol}^{-1}$

$\Delta H_f(CO) = -137 \text{ kJ mol}^{-1}$

$\Delta H_f(CO_2) = -394.5 \text{ kJ mol}^{-1}$

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

$\Delta Hr = ?$

$\Delta Hr = \Sigma(\Delta H_f)_{products} - \Sigma(\Delta H_f)_{reactants}$

$\Delta Hr = [2\,\Delta H_f(Fe) + 3\Delta H_f(CO_2)] - [\Delta H_f(Fe_2O_3) + 3\Delta H_f(CO)]$

$\Delta Hr = [0 + 3(-394.5)] - [-741 + 3(-137)]$

$\Delta Hr = [-1183.5] - [-1152]$

$\Delta Hr = -1183.5 + 1152$

$\Delta Hr = -31.5 \text{ kJ mol}^{-1}$

66.

Solution:

Given :

$T = 175^0\,C = 175 + 273 = 448K$

Concentration of 1-pentyne

[A] = 1.3%

Concentration of 2-pentyne

[B] = 95.2%

Concentration of 1, 2-pentadiene

[C] = 3.5%

At equilibrium

$B \rightleftharpoons A$

95.2% 1.3% $\Rightarrow$

$K_1 = \frac{1.3}{95.2} = 0.0136$

$B \rightleftharpoons C$

95.2% 3.5% $\Rightarrow$

$K_2 = \frac{3.5}{95.2} = 0.0367$

$\Rightarrow \Delta G_1^0 = -2.303\,RT\log K_1$

$\Delta G_1^0 = -2.303 \times 8.314 \times 448 \times \log 0.0136$

$\Delta G_1^0 = +16010 \text{ J}$

$\Delta G_1^0 = +16 \text{ kJ}$

$\Rightarrow \Delta G_2^0 = -2.303\,RT \log K_2$

$\Delta G_2^0 = -2.303 \times 8.314 \times 448$